

WYVERN-E — Research Proposal

A Skylight Rocketry Venture · 70 mm Single-Stage Servo-TVC Sustainer

Allison Hong · Chris Liu · Swaroop K. Sahoo

Executive Summary

WYVERN-E is a 70 mm, single-stage, **3D-printed active-thrust-vector-control (TVC)** research rocket that demonstrates closed-loop flight stabilization on a bare-metal **Raspberry Pi Pico 2 W (RP2350)** flight computer, powered by the **Estes F15-4**. It carries four 72 mm fins (no ballast) for **passive stability during launch and the F15 ignition spike**, after which ($t \geq 0.5$ s) the TVC loop engages on the smooth portion of the thrust curve to stabilize and execute a commanded maneuver. The two candidate TVC actuators — a tri-solenoid *magnetic* gimbal and a *servo* gimbal — are compared quantitatively on a purpose-built **3-axis thrust-vector balance** that resolves both thrust magnitude and vector direction; the flight vehicle carries the servo system. The program retains the static motor/materials test regimen; the aerofoil wind-tunnel campaign was removed in 2026-08 and its aerodynamic question is now answered analytically and validated in flight (RQ3). All flights are FAA Class-1 (no waiver, no certification). Predicted apogee ≈ 324 ft; project cost $\approx \$1,720$.

1. Background & Motivation

1.1 Thrust vector control

Thrust vectoring steers a rocket by gimbaling the exhaust, producing a control moment without aerodynamic surfaces — the method used by every orbital launch vehicle. A small, low-cost, fully printed TVC vehicle makes the control problem tangible while producing real flight data.

1.2 WYVERN lineage

WYVERN-E 1.0 (interceptor study) → 2.0 (84 mm two-stage, custom avionics) → 3.0 (Raspberry Pi 5, solenoid-vs-servo A/B in flight) established the aero, structures, and avionics groundwork. 4.0 distills those lessons into the simplest vehicle that still answers the core question: a single stage, a single bare-metal controller, and the A/B actuator comparison moved to a repeatable ground balance.

1.3 The 4.0 thesis

A small finned rocket can use passive stability to survive launch and the ignition transient, then hand authority to a closed-loop TVC system on the smooth thrust curve — and the relative merit of magnetic vs servo gimbals can be measured directly on the ground before flight.

2. Research Questions

Numbering matches WYVERN_E4_Proposal_rev2.md §2 (canonical). The wind tunnel and the airfoil-CFD package were removed from the program in 2026-08; the aerofoil-polar question they served is replaced by RQ3 below, which is answered analytically and validated in flight.

#	Question	Method A	Method B	Primary metric
RQ1	Magnetic vs servo TVC actuation	3-axis thrust-vector balance, F15-0	bench signal model + SIL	bandwidth, slew, overshoot, SSE, max vector angle
RQ2	PLA vs PETG-CF for TVC structures	3-point bend coupons, both materials	lumped-capacitance thermal + measured engine-bay wall temp from static fires	flexural modulus, HDT margin
RQ3	Predicted vs in-situ passive stability	Barrowman CP/CN + RK4 dispersion	flight-telemetry reconstruction of margin and coast Cd	Δ static margin (cal), Δ Cd
RQ4	Closed-loop gain sensitivity in flight	24-point phase/gain-margin sweep + Monte Carlo	onboard log, up to 4 flights	pitch error, gimbal track, PM/GM, recovery

3. Vehicle Architecture

3.1 Configuration & mass budget

70 mm OD, 0.74 m, three bays + two bulkheads; 4 fins (72 mm), no ballast.

Section	Key mass	Subtotal
Nose (PLA)	ellipsoid nose	21 g
Recovery bay (PLA)	bay tube, chute+Kevlar cord, Nomex, BNO085 (vote), PETG-CF bypass tube, PETG-CF sealed Bulkhead B, ejection plenum	137 g
FC bay (PLA)	bay tube, Pico 2 W, BNO085, baro, μ SD, i3 4K Thumb	122 g

	Action Camera, 2S LiPo + 5 V UBEC	
Engine/TVC bay (PETG-CF)	bay tube, Bulkhead A, gimbal, 2 servos, BN0085, mount	268 g
Structure	4 PLA fins (72 mm) + wiring	50 g
Dry		690 g
Motor	F15-4 (60 g prop)	102 g
Liftoff		792 g

3.2 Materials

- **PETG-CF** ($\rho \approx 1.30 \text{ g/cm}^3$, HDT $\approx 80 \text{ }^\circ\text{C}$): both bulkheads, the ejection bypass tube, and the engine assembly (engine/TVC bay + motor mount + gimbal) — i.e. the whole ejection-gas path plus the TVC structures.
- **PLA** ($\rho \approx 1.24 \text{ g/cm}^3$, HDT $\approx 55 \text{ }^\circ\text{C}$): nose, both body tubes, fins — everything with no motor heat and no gas contact. Walls run 1.2 mm (3 perimeters) rather than 1.6 mm; the airframe is print/handling-limited at $\sim 340\times$ safety factor, so the thicker wall was carrying margin it never needed, and dropping it recovers 45.6 g.
- **Zoning rationale is thermal, not mass.** PLA at $55 \text{ }^\circ\text{C}$ HDT cannot be trusted in contact with ejection gas; PETG-CF at $80 \text{ }^\circ\text{C}$ is the minimum defensible material for that path.

3.3 Stability — fins + ballast + the 0.5 s rule

An apogee sweep shows ballast lowers altitude, so we use **no ballast** and size fins to the minimum stable 1.0 cal: $4 \times 72 \text{ mm fins} \rightarrow \text{CP } 56.8 \text{ cm, CG } 48.4 \text{ cm} = \mathbf{+1.20 \text{ cal}}$ static margin (stable). This passive margin holds the vehicle through launch and the F15 ignition spike; the TVC controller is **inhibited until $t = 0.5 \text{ s}$** , then engages on the smooth curve. A historical finless variant (margin -5.6 cal) was rejected because it is statically unstable and cannot survive the pre-TVC transient.

3.4 Structural & thermal margins

First-order analysis (we4_analysis.py): minimum safety factor **$> 300\times$** (the 25 N motor leaves the 1.6 mm airframe print/handling-limited, not load-limited); fin flutter velocity well above the 25 m/s flight regime; engine-bay thermal margin to PETG-CF HDT $\approx 40 \text{ }^\circ\text{C}$ (wall peaks $\sim 40 \text{ }^\circ\text{C}$ against an $80 \text{ }^\circ\text{C}$ HDT).

4. Propulsion & Trajectory

4.1 Motors (verified)

Motor	Spec	Role	Qty
Estes F15-4	49.6 N·s, 14.4 N	flight only (ejection	4

	avg / 25.3 N pk, 3.45 s, 4 s delay + ejection	= recovery)	
Estes F15-0	49.6 N·s, 14.4 N avg / 25.3 N pk, 3.45 s, 0-delay (plugged)	static stand + TVC balance (MTVC and servo)	10
Estes E16-4	~ 16 N avg	stand commissioning	6

4.2 Predicted performance (unified RK4 + Barrowman, `we4_flightsim.py`)

T/W 1.85 avg / 3.26 peak; Cd 0.539; burnout 3.45 s, 59.1 m, 28.9 m/s; **apogee ~324 ft @ 6.27 s**; motor ejection at $t = 7.45$ s (+1.18 s past apogee) @ 11.5 m/s; 24" chute → 5.0 m/s descent. Dispersion (± 5 % mass, ± 15 % Cd): see plots4/06_dispersion.png.

5. Flight Computer & Control

5.1 Raspberry Pi Pico 2 W (RP2350)

A single dual-core 150 MHz RP2350 is flight computer *and* 500 Hz controller: core 0 reads three BNO085 + two baros (BME688 + BMP388) and closes the TVC loop driving 2 servos via hardware PWM; core 1 handles microSD logging and Wi-Fi bench telemetry. No Linux, no scheduler jitter. Recovery is the motor's own ejection charge (the FC only logs/observes).

5.2 IMUs — Game Rotation Vector

All three BNO085 run in **Game Rotation Vector** (accel + gyro, magnetometer disabled) to reject the magnetic interference of the adjacent servos. Gimbal deflection = $q_{body}^{-1} \otimes q_{gimbal}$, giving true nozzle attitude relative to the body (catches linkage backlash/flex). The recovery-bay unit votes against the FC unit for fault detection.

5.3 Control law

Per-axis PID ($K_p = 0.10$, $K_i = 0.40$, $K_d = 0.18$; margin- and auto-tune-validated), output clamped to $\pm 8^\circ$, servo lag $\tau \approx 0.04$ s. **TVC inhibited for the first 0.5 s** (ignition spike) — fins hold attitude — then engages: stabilize to vertical, then a 4° commanded maneuver. Required gimbal torque 0.56 kg·cm (micro-servo class); control authority positive throughout the powered phase.

5.4 Power & data

A light 2S LiPo (7.4 V, ~450 mAh) feeds a single 5 V/6 V UBEC (set 5 V) whose one rail powers Pico 2 W VSYS, the camera, and both servos (EMAX ES08MA II, running at 5 V ≈ 1.8 kg·cm). Shared-rail decoupling (1000 μ F bulk at the servos, 100 μ F + an SS34 hold-up Schottky at VSYS) keeps servo-stall transients from browning-out the Pico. No recovery battery — recovery is the motor's own ejection charge. Pack voltage is monitored on GP26/ADC0 (100k/62k divider; warn 6.4 V, arm-inhibit 6.0 V). Onboard log: full-rate

IMU/baro/control + i3 4K Thumb Action Camera 1080p60 video. The power+camera group (LiPo ~30 g + UBEC ~10 g + i3 4K Thumb Action Camera ~36 g $\approx$ 76 g) sits inside the 122 g FC-bay budget.

6. Recovery

Recovery uses the **F15-4 motor's own ejection charge** (4 s delay $\rightarrow$ fires $t \approx 7.45$ s, ~ 0.66 s past apogee), routed through a solid-walled 12 mm PETG-CF bypass tube past the *sealed* FC bay into the recovery bay to release a friction-fit nose — **no RRC3+, no 9 V, no e-match/BP, no CO2, no FC involvement**. Feasibility (we4_ejection_feasibility.py): tube loss ≈ 0.06 kPa; bay pressurizes to ~ 140 kPa vs a 14–41 kPa nose-release threshold = **3.4 $\times$ margin**. F15-4 is the closest Estes delay to the ~ 3.5 s coast optimum (F15-6/-8 eject 2.5 s/4.5 s late, too fast/low). Single passive event, no electronic backup. Opening at ~ 11.5 m/s; 1/8" Kevlar cord ($> 800\times$ margin) + Nomex protector; 24" chute $\rightarrow \sim 6$ m/s.

7. Ground Test Program

7.1 3-axis thrust-vector balance

Motor + gimbal on a thrust block restrained by one axial (5 kg) + two lateral (1 kg) strain-gauge load cells $\rightarrow T, \theta, \phi$. Actuator-agnostic — both magnetic and servo systems tested identically. RQ3 metrics logged vs commanded. ### 7.2 Static thrust + materials stand Axial cell + steel deflector: validates the F15-0 thrust curve and carries the engine-bay wall thermocouple for RQ2 (plugged 0-delay on the ground — no ejection into the fixture). Jetvane screening is out of scope as of 2026-08. ### 7.3 Motor & calibration plan Load cells dead-weight calibrated, then commissioned with **6 $\times$ E16-4** (3 per stand). Counts: F15-4 $\times 4$ (flight), F15-0 $\times 13$ (ground), E16-4 $\times 6$.

8. Simulation Suite (RQ3/RQ4 Method A)

RK4 + Barrowman trajectory with componentwise drag buildup, power-law wind shear, and Monte Carlo dispersion over the atmospheric and build-tolerance envelope; 500 Hz closed-loop pitch model with servo lag, transport delay, and Dryden-spectrum gust forcing; phase/gain-margin sweep across the burn-time $\times$ atmosphere grid; and a software-in-the-loop flight computer writing logs in the onboard recorder's schema. See Simulations/README.md.

9. Safety & Regulatory

Single F15-4: 49.6 N·s, 60 g propellant, $\leq$ F class, liftoff 792 g < 1500 g $\rightarrow$ **FAA Class-1, no waiver, no Level-1 certification**. Remote ignition, ≥ 3 m standoff on the stands, gimbal-neutral fail-safe, motor-integral ejection (igniter installed at the pad; no electronic ejection circuit to arm or inhibit).

10. Budget

$\approx$ **\$1,720** total program spend (vehicle + 3-axis balance + static/materials stand + one-time tools + all motors): \$1,241 still to buy + \$479 already acquired. This is down from the

\$1,882 originally scoped; the Hofferth wind tunnel section was deleted from the BOM in the 2026-08 scope change. Live per-line pricing in Documentation/WYVERN_E4_BOM.xlsx. Per-flight consumable $\approx$ F15-4 \$17 (integral delay/ejection; no separate initiator or BP charge).

11. Schedule & Milestones

Phase	Wk	Milestone
Print + assemble	1–3	airframe, gimbal, both stands
Bench bring-up	4	self-test all-PASS, control loop dry-run
TVC balance A/B (RQ1)	5–6	magnetic vs servo dataset
Static fires (RQ2)	7	thrust-curve verification + engine-bay wall temperature
Sim + margin analysis (RQ3/RQ4)	8–9	dispersion, PM/GM sweep, SIL
Flight tests (RQ3/RQ4)	10–11	up to 4 flights, onboard logs
Analysis + paper	12–14	results, paper

12. Risk Register

Risk	Likelihood	Mitigation
Pre-TVC instability	Med	4× 72 mm fins, no ballast (+1.20 cal at liftoff); TVC inhibit 0.5 s
Servo slew too slow	Med	bench-verify on balance before flight; fast digital micro
Hard deploy at ~ 11.5 m/s (motor ejection, +1.18 s past apogee)	Low	Kevlar harness >800× margin, Nomex protector, ground-tested charge
Launch-detect miss at low T/W	Low	tune arming alt; verify on 2.2 g spike
Camera/SD throughput	Low	self-contained i3 4K Thumb Action Camera (decoupled from FC)

13. Expected Outcomes & Deliverables

A flight-validated small TVC vehicle; a quantitative magnetic-vs-servo TVC dataset; a quantified accuracy bound on Barrowman-class stability prediction versus flight telemetry; a PLA-vs-PETG-CF structural comparison justifying the flown material zoning; full open documentation (CAD, firmware, wiring, sims) and a research paper.

References

Barrowman, J. *The Practical Calculation of the Aerodynamic Characteristics of Slender Finned Vehicles*, NASA, 1967. Box, J. (BPS.space) *Signal/Echo* TVC flight series. NAR/Tripoli Model Rocket Safety Codes. Bosch Sensortec BNO085 datasheet; Raspberry Pi RP2350 / Pico 2 W datasheet.